

Technology Stack

Date	15 JULY 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML5, CSS3, Bootstrap	Provides a responsive and user-friendly interface for borrowers to manage loans and settlements.

2	Backend / Server-Side	Python (Flask)	Handles authentication, AI processing, settlement prediction, and API services efficiently.
3	Database / Data Storage	PostgreSQL	Stores Users, Loans, Financial Profiles, Settlement Records, and AI History with strong relational support and normalization.
4	Cloud / Hosting / Deployment	AWS / Render / Docker	Provides scalable deployment, secure hosting, and easy maintenance.
5	Version Control & CI/CD	GitHub, GitHub Actions	Enables collaborative development, version tracking, and automated deployment.

6	Third-Party APIs / Other Tools	OpenAI API, Firebase Authentication	Generates AI negotiation strategies, settlement letters, and secure user authentication.